

Lab Report On Installation of Kali Linux on VMware Workstation

Objective

The objective of this experiment is to study the process of installing the Kali Linux operating system on a virtual machine using VMware Workstation Player and to understand the concept of operating system virtualization.

Introduction

Kali Linux is a Debian-based Linux distribution mainly used for cybersecurity, ethical hacking, and penetration testing. Installing Kali Linux directly on physical hardware may affect the existing operating system. Virtualization solves this problem by allowing multiple operating systems to run on the same physical machine. VMware Workstation is a popular virtualization software that enables users to create and manage virtual machines safely and efficiently.

Software and Hardware Requirements

Software Requirements:

- VMware Workstation Player
- Kali Linux ISO image
- Windows 10/11 (Host Operating System)

Hardware Requirements:

- Minimum 8 GB RAM (4 GB recommended for VM)
- At least 20 GB free disk space
- Intel/AMD processor with virtualization support enabled

Theory

Virtualization is a technology that allows one physical computer to run multiple operating systems simultaneously. VMware Workstation creates a virtual environment where a guest operating system, such as Kali Linux, can run independently of the host operating system. Kali Linux provides a wide range of security tools used for network testing, vulnerability analysis, and digital forensics, making it an important operating system for cybersecurity education.

Procedures

1. VMware Workstation Player was installed successfully on the host computer.
2. The Kali Linux ISO file was downloaded from the official Kali Linux website.
3. VMware Workstation was launched and the option **Create a New Virtual Machine** was selected.
4. The **Installer disc image file (ISO)** option was chosen and the Kali Linux ISO file was selected.
5. The guest operating system was set to **Linux** and version to **Debian (64-bit)**.
6. A name and location for the virtual machine were provided.

7. Hardware resources were allocated, including:
 - RAM size
 - Number of processor cores
 - Virtual hard disk size
8. The virtual machine was powered on and the Kali Linux boot menu appeared.
9. **Graphical Install** option was selected and the installation steps such as language, location, keyboard layout, and user account were configured.
10. After completion of the installation, the system was rebooted.
11. Kali Linux started successfully and the login screen appeared.

Observation

- Kali Linux booted correctly inside the virtual machine.
- All basic system functions were working properly.
- The graphical user interface loaded successfully.

Result

Kali Linux was successfully installed and executed in VMware Workstation Player. The virtual machine provided a stable and secure environment for running Kali Linux.

Conclusion

This experiment demonstrated the successful installation of Kali Linux using VMware Workstation. Virtualization allows users to explore and use advanced operating systems like Kali Linux without risking the host system. This method is efficient, safe, and suitable for learning cybersecurity tools and techniques.